Chemical-free microwave seed treatment

Parostok treats seed before sowing with a controlled 2.45 GHz microwave field. Dry process, no chemicals, no residues on the seed.



New design (rendering). The specifications refer to the current production model.

How it works

- Seed flows continuously through a controlled microwave field.
- The dose is set by exposure time at fixed power; each crop and variety has its own mode.
- Treatment can be offered as a service or with the machine installed on site.

Key specifications (current model)

Power supply	**110 / 220 V**
Power consumption	**3.0–6.0 kW**
Throughput	**0.4–2.0 t/h**
Weight	**210 kg**
Dimensions	**70 × 90 × 206 cm**
Loading	**from the top**
Microwave frequency	**2.45 GHz (ISM band)**

Evidence to date

- **Independent laboratory, April 2026** (Plant Health Institute, Ukravit Science Park; DSTU 4138-2002): 36 corn seed samples from 6 lots, 5 modes — uniform root development, no signs of suppression.
- **Weaker seed gains most:** in the same test the lowest lot (86 % germination) gained 6.2 points, while lots at 90–92 % changed by −1.6 to +1.6 points.
- **Field, Poland 2024** (RSUW Głuchów, KWS Nataelo): 11.97 vs 10.34 t/ha (+15.7 %). Single plots without replication — indicative only.
- **Research since 2000** with Ukrainian research institutes: 24 crops in the 2000–2001 programme, replicated field trials on winter wheat.

For whom

Seed producers and processors, cooperatives and large farms — especially for seed lots below their germination standard.

PE SPE “KARE” · Odesa, Ukraine · mwparostok@gmail.com · www.parostokmw.com

October 2026. Figures are the company’s own unless marked as independent.